Differentiation: Inverse Trigonometric Functions

Questions 1–26

Question 1. Show that

a) $\dfrac{d}{dx}(\arcsin x) = \dfrac{1}{\sqrt{1-x^2}}$

b) $\dfrac{d}{dx}(\arccos x) = -\dfrac{1}{\sqrt{1-x^2}}$

c) $\dfrac{d}{dx}(\arctan x) = \dfrac{1}{1+x^2}$

Question 2. Find the gradient of the tangent to the curve:

a) $y = \arcsin x$ at $x = \frac{1}{3}$

b) $y = \arccos x$ at $x = -\frac{1}{4}$

c) $y = \arctan x$ at $x = \frac{2}{5}$

The following questions use the derivatives of $\arcsin kx$, $\arccos kx$ and $\arctan kx$ and require no further Chain, Product or Quotient Rule.

Question 3. Differentiate each of the following:

a) $y = \arcsin 2x$

b) $y = \arccos 3x$

c) $y = \arctan 8x$

d) $y = \arcsin \dfrac{x}{3}$

e) $y = \arccos \dfrac{x}{5}$

f) $y = \arctan\left(-\dfrac{2}{3}x\right)$

Question 4. Show that

a) $\dfrac{d}{dx}\left(\arcsin \dfrac{x}{2}\right) = \dfrac{1}{\sqrt{4-x^2}}$

b) $\dfrac{d}{dx}\left(\arccos \dfrac{3x}{4}\right) = -\dfrac{3}{\sqrt{16-9x^2}}$

c) $\dfrac{d}{dx}\left(\arctan \dfrac{5x}{3}\right) = \dfrac{15}{9+25x^2}$

Question 5. Find the exact gradient of the curve $y = 8\arcsin\dfrac{x}{4}$ at the point with x-coordinate 2.

Question 6. Find the equation of the tangent to the curve $y = 3\arccos x$ when $x = \frac{1}{2}$, writing your answer in the form $y = ax + b$ where a and b are given in exact form.

Question 7. Find the equation of the tangent to the curve $y = 2\arccos\dfrac{x}{3} - \arcsin\dfrac{x}{3}$ when $x = \dfrac{3}{\sqrt{2}}$, writing your answer in the form $y = ax + b$ where a and b are given in exact form.

Question 8. Find the equation of the tangent to the curve $y = 4\arctan\left(\dfrac{2x}{5}\right)$ when $x = \dfrac{5}{2}$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

Question 9. Find the equation of the normal to the curve $y = 4 - 3\arctan\dfrac{x}{4} - 2\arctan\dfrac{x}{3}$ when $x = 0$, writing your answer in the form $px + qy = r$ where p, q and r are integers.

Question 10. Given $f(x) = \arctan x$ and $g(x) = \arctan 3x$, solve the equation $f'(x) = g'(x)$.

Question 11. A curve has the equation $y = \arcsin\left(\frac{2}{3}x\right)$, where $-\frac{3}{2} \leq x \leq \frac{3}{2}$.

a) Show that the gradient of the curve is never equal to $\frac{1}{2}$.

The point P lies on the curve with x-coordinate $-\frac{3\sqrt{2}}{4}$. The normal to the curve at P passes through the y-axis at the point Q.

b) Show that the exact area of the triangle OPQ, where O is the origin, can be written as $\frac{3\sqrt{2}}{64}(9 + 2\pi)$ units^2.

Question 12. Show that the curve with equation $y = \arcsin x - \arccos x$ is increasing for $-1 < x < 1$.

Question 13. It is given that $y = 3\arctan 6x$.

a) Show that there are no points on the curve where the gradient is equal to 24.

b) State the maximum gradient of the curve.

Question 14. Find the exact coordinates of the single stationary point of the curve $y = \arctan 2x - x^2$. **Prerequisite knowledge for Q15: Binomial Series**

Question 15. It is given that $f(x) = \arctan 3x$.

a) Find $f'(x)$.

b) Assuming that x is small, use your answer to part (a) to express $f'(x)$ in ascending powers of x as far as the term in x^4.

c) Hence express $\arctan 3x$ in ascending powers of x as far as the term in x^5.

Question 16. It is given that $f(x) = p\arcsin x - qx$, where p and q are positive constants. Show that the curve $y = f(x)$ has precisely two stationary points when $p < q$.

Question 17. Find the values of x for which the curve with equation $y = 5x + \sqrt{21}\arccos x$ is increasing. Write your answer in set notation.

Question 18. Show that the curve with equation $y = 3\arctan\dfrac{x}{6} - 2\arctan\dfrac{x}{4}$ has a stationary point of inflection at the origin. **The following questions use the derivatives of** $\arcsin kx$**,** $\arccos kx$ **and** $\arctan kx$ **and use any differentiation knowledge from A-Level Maths.**

Question 19. Differentiate each of the following:

a) $y = \arcsin x^2$

b) $y = \arccos \sqrt{x}$

c) $y = \arctan \dfrac{1}{x}$

d) $y = \cos(\arcsin 2x)$ for $-\dfrac{1}{2} < x < \dfrac{1}{2}$

e) $y = \sin\left(\arccos \dfrac{x}{2}\right)$ for $-2 < x < 2$

f) $y = \cos(\arctan 4x)$ for $x \in \mathbb{R}$

Question 20. Find the equation of the tangent to the curve $y = (\arcsin x)^2$ when $x = \dfrac{\sqrt{3}}{2}$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

Question 21. Find the equation of the tangent to the curve $y = \arctan(e^x)$ when $x = -\frac{1}{2}\ln 3$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

Question 22. Differentiate each of the following:

a) $y = x \arcsin x$

b) $y = x^2 \arccos 2x$

c) $y = x^3 \arctan(x^2)$

Question 23. It is given that $y = (x-5)\sqrt{-x^2+10x-24} + \sin^{-1}(x-5)$, where $4 \leq x \leq 6$. Show that $\dfrac{dy}{dx} = k\sqrt{-x^2+10x-24}$, where k is an integer to be found.

Question 24. It is given that

$$y = 3x\sqrt{4-x^2} + 12\tan^{-1}\left(\frac{x}{\sqrt{4-x^2}}\right), \qquad -2 < x < 2.$$

Show that $\dfrac{dy}{dx} = k\sqrt{4-x^2}$, where k is an integer to be found.

Question 25. It is given that a curve has equation $\arccos y - \arccos 3x = \dfrac{\pi}{3}$, where $-\dfrac{1}{6} \leq x \leq \dfrac{1}{3}$. Find the value of $\dfrac{dy}{dx}$ when $x = \dfrac{1}{6}$.

Question 26. Show that

a) $\dfrac{d}{dx}(\text{cosec}^{-1} x) = -\dfrac{1}{\sqrt{x^4 - x^2}}$

b) $\dfrac{d}{dx}(\sec^{-1} x) = \dfrac{1}{\sqrt{x^4 - x^2}}$

c) $\dfrac{d}{dx}(\cot^{-1} x) = -\dfrac{1}{1 + x^2}$